AI-Assisted Mathematical Reasoning and Statistical Outlier Detection

Independent Mathematics & AI Project — 2026

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Project type: Independent academic/technical project

Academic Integrity Statement

This project was created independently in 2026. It was not completed as a university project during the bachelor's degree and is not presented as a peer-reviewed publication. The dataset is synthetic and created for educational demonstration.

Abstract

This project investigates the use of classical statistical reasoning and machine-learning-based anomaly detection to identify unusual observations in a synthetic quantitative dataset. The workflow combines descriptive statistics, standardized-score screening, Isolation Forest, visualization, and human verification. A second objective is to demonstrate a practical human-in-the-loop approach relevant to mathematical AI evaluation: computational tools can generate and test evidence, but the final assessment requires independent reasoning about assumptions, calculations, and context.

1. Introduction

Modern AI systems increasingly generate mathematical explanations, quantitative estimates, and multi-step reasoning. A useful mathematics expert workflow is therefore not limited to obtaining an answer; it also requires checking whether the reasoning is valid, the calculations are correct, and unusual cases have been handled appropriately. This project connects the author's Mathematical Sciences background with current AI evaluation practices by using a small reproducible quantitative experiment.

Outlier AI's current mathematics opportunity describes work involving complex problem sets, mathematical questions, reasoning traces, and ranking responses for scientific rigor. This project is intentionally structured to demonstrate related analytical habits without claiming prior professional AI-training employment.

2. Objectives

• Apply descriptive statistics to quantitative variables.

• Detect unusual observations using standardized scores.

• Compare a classical statistical screen with Isolation Forest.

• Use visualization to inspect relationships and flagged observations.

• Demonstrate independent verification rather than accepting computational output automatically.

• Create a reproducible portfolio project relevant to mathematics and AI evaluation.

3. Dataset Design

The dataset contains 120 synthetic records. Three variables were generated: weekly study hours, practice score, and solution accuracy. Six records were deliberately modified to create unusual combinations. No real people or confidential data are represented.

Variable	Description	Role
study_hours	Simulated weekly study time	Quantitative
practice_score	Simulated score from 40–100	Quantitative
accuracy	Simulated solution accuracy	Proportion
record_id	Synthetic identifier	Identifier

4. Mathematical Methodology

For each variable, the standardized score is calculated as:

$$z = (x - \mu) / \sigma$$

where x is an observation, μ is the mean, and σ is the standard deviation used in the screening calculation. The maximum absolute standardized score across the three variables is used as a simple multivariate screening signal. A threshold of 2.5 was selected for this educational experiment.

This threshold is a screening choice, not a universal definition of an outlier. Real-world interpretation depends on the data-generating process and domain context.

5. Machine-Learning Method: Isolation Forest

Isolation Forest is an unsupervised anomaly-detection method based on random recursive partitioning. Observations that are isolated with shorter paths are more likely to be anomalous. The implementation uses 200 estimators, a contamination setting of 0.05, and a fixed random seed for reproducibility.

The project uses Isolation Forest as a complementary signal rather than treating it as an infallible classifier.

6. Results

The experiment intentionally contains 6 ground-truth synthetic anomalies. The combined statistical/Isolation Forest screen flagged 6 records.

Metric	Value
True positives	6
False positives	0
False negatives	0
True negatives	114
Precision	100.00%
Recall	100.00%
F1 score	100.00%

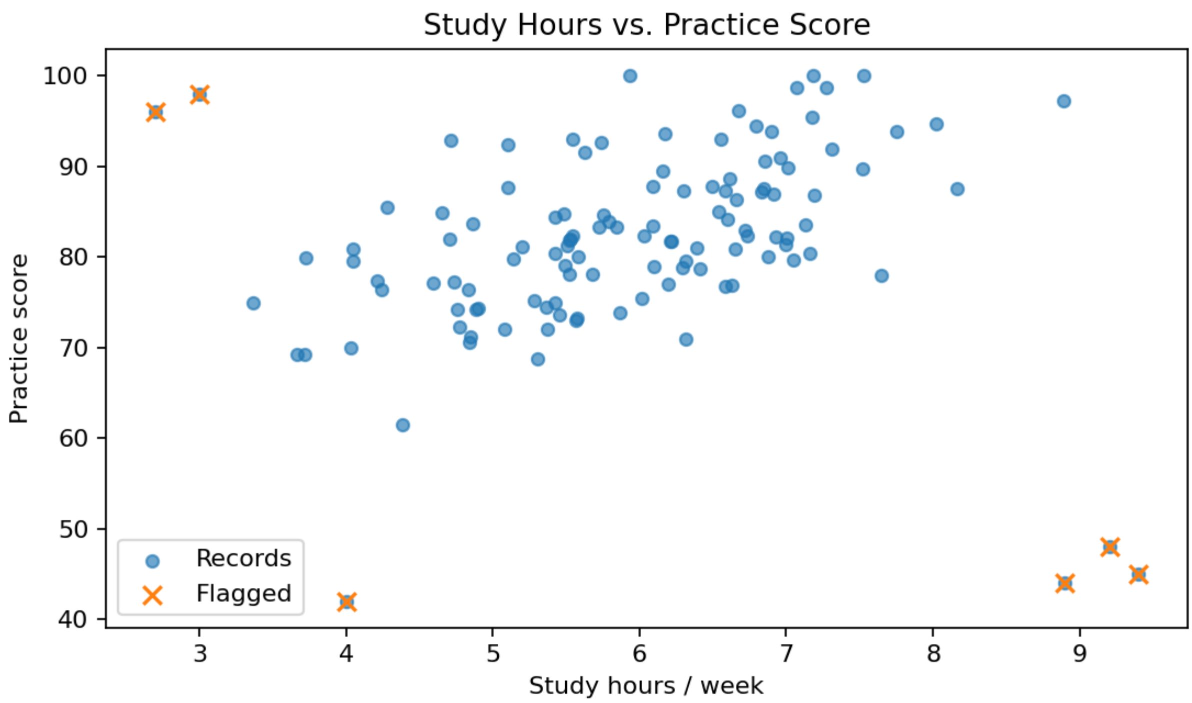

Figure 1. Relationship between study hours and practice score; crosses indicate records flagged by the combined screening method.

7. Statistical Profile

Variable	Mean	Median	Std. Dev.	Minimum	Maximum
study_hours	5.941	5.975	1.247	2.700	9.400
practice_score	81.693	81.882	10.534	42.000	100.000
accuracy	0.976	0.990	0.081	0.460	0.990

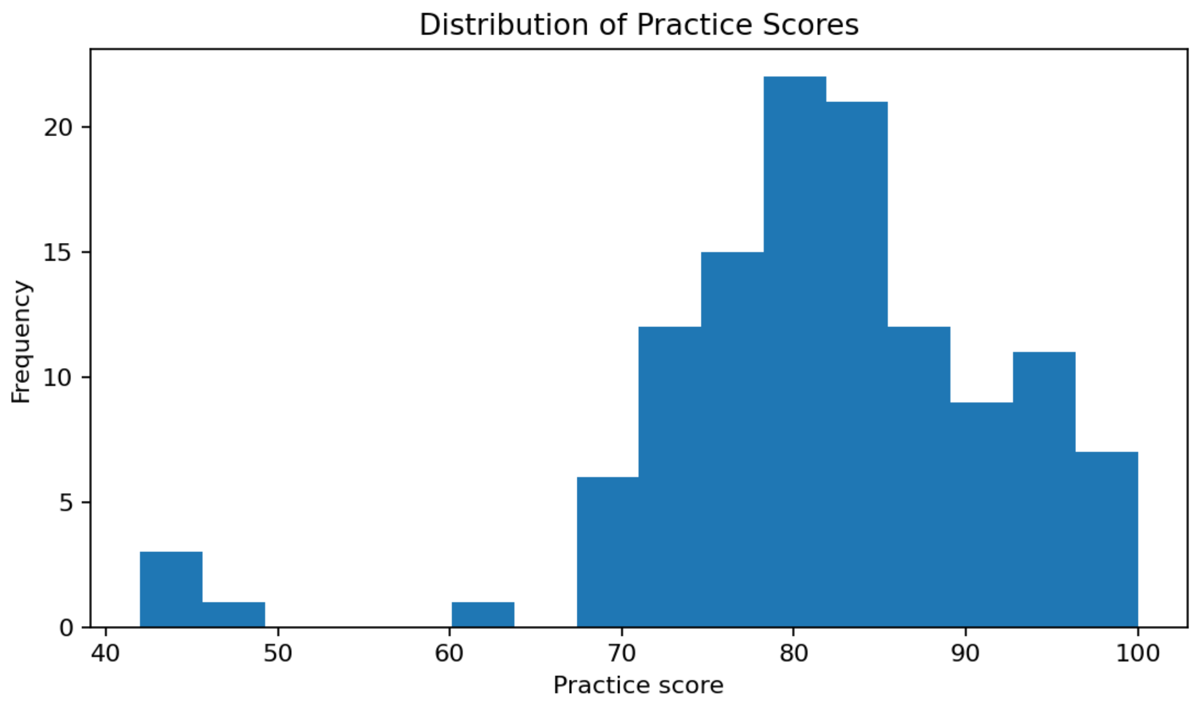

Figure 2. Distribution of synthetic practice scores.

8. Comparing Statistical and ML Signals

The two methods answer slightly different questions. A z-score asks how far an observation lies from a variable's central tendency in standardized units. Isolation Forest instead uses the structure of the multivariate feature space to identify observations that can be isolated unusually quickly. Agreement between methods can increase confidence in a screening result, while disagreement is a reason for closer human review.

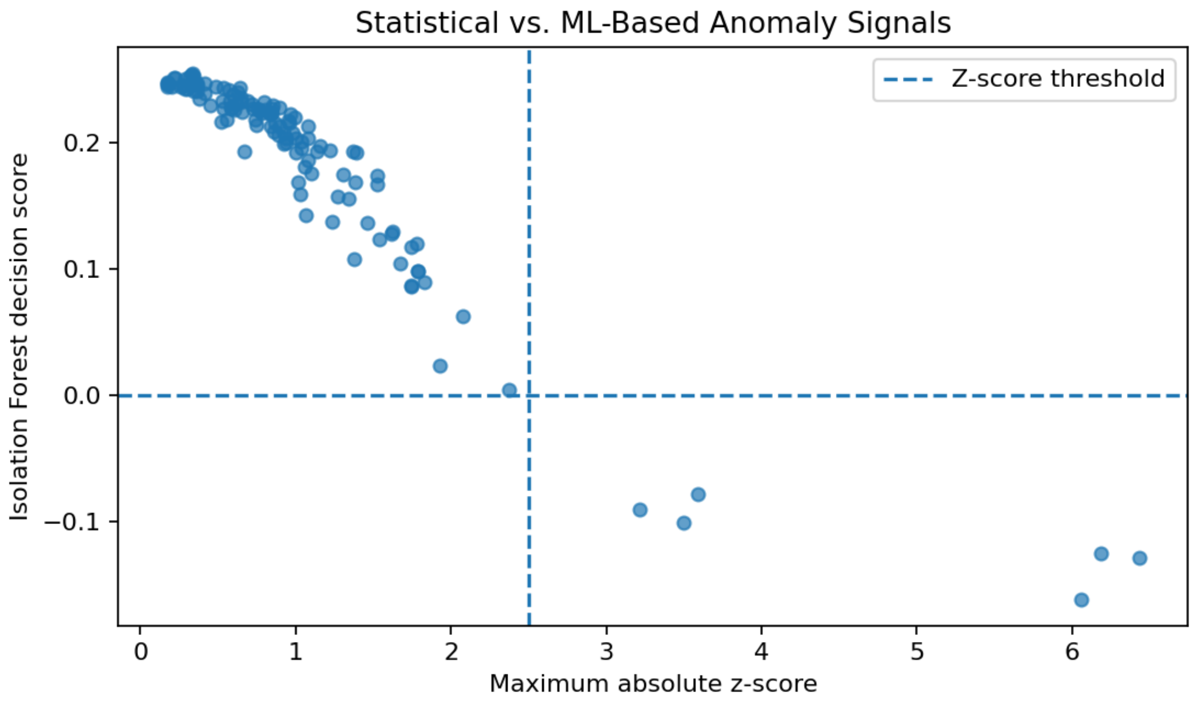

Figure 3. Comparison of standardized-score magnitude and Isolation Forest decision score.

9. AI-Assisted Mathematical Reasoning Workflow

Stage	Human / AI role	Quality check
1. Prompt	Define a precise quantitative problem	Ambiguity and assumptions
2. Generate	AI proposes a solution or alternative methods	Do not assume correctness
3. Derive	Human independently solves the problem	Formula and logic
4. Compute	Python/Excel checks arithmetic	Numerical consistency
5. Evaluate	Compare AI and independent solutions	Correctness, rigor, completeness
6. Rank	Assign a rubric-based quality score	Evidence-based decision

10. Example Evaluation Rubric

Criterion	Weight	What is checked
Mathematical correctness	35%	Correct formulas, calculations and conclusion
Logical reasoning	25%	Valid step-by-step argument
Completeness	15%	All required parts and assumptions
Clarity	15%	Readable and unambiguous explanation
Efficiency	10%	Appropriate method without unnecessary work

11. Discussion

The experiment demonstrates why mathematical expertise remains useful in AI evaluation. An AI system may produce a plausible-looking explanation while making a small but consequential mathematical error. Independent derivation and computational verification create a second line of defense. Likewise, an anomaly detector can flag a record without proving that the record is wrong. Human reasoning is needed to distinguish a legitimate rare case from a data-quality problem.

12. Limitations

• The dataset is synthetic and intentionally designed for demonstration.

• The six anomalies were manually injected, so the benchmark does not represent a naturally occurring data problem.

• A 2.5 z-score threshold and 5% Isolation Forest contamination setting are project choices rather than universal standards.

• Outlier detection does not establish causality or prove an observation is erroneous.

• The AI-evaluation workflow is conceptual and does not claim professional employment or prior Outlier project participation.

13. Future Work

• Compare Isolation Forest with Local Outlier Factor and robust statistical methods.

• Build a benchmark of difficult mathematical prompts and verified reference answers.

• Evaluate multiple LLM responses using the proposed rubric.

• Add symbolic mathematics checks and LaTeX-based proof validation.

• Study error patterns such as arithmetic mistakes, invalid assumptions, circular reasoning, and unsupported conclusions.

14. Conclusion

This project combines mathematical reasoning, statistics, computing, and AI-oriented evaluation into a reproducible independent study. It demonstrates a practical principle for mathematical AI work: generate or inspect a solution, verify it independently, identify errors or unusual cases, and document the evidence behind the final judgment. The project is intended as a portfolio artifact for demonstrating analytical capability and learning, not as a claim of peer-reviewed research.

15. References

• scikit-learn Developers. Isolation Forest documentation and anomaly-detection examples. https://scikit-learn.org/

• Outlier AI. Math Expertise for AI Training — current opportunity description. https://outlier.ai/math/abc0

• Outlier AI. AI trainer overview and common expert tasks. https://outlier.ai/